

UEC716: DATABASE MANAGEMENT SYSTEMS

L	T	P	Cr
3	0	2	4.0

Course Objectives: To provide students with a strong foundation in the principles, design, and implementation of database systems, while understanding the underlying concepts of database architecture, normalization, transactions, and query optimization.

Introduction: Data, data processing requirement, desirable characteristics of an ideal data processing system, traditional file based system & its drawback, Database system architecture: Data Abstraction, Data Independence, Data Definition and Data Manipulation Languages.

Data models: Entity-relationship, network, relational and object oriented data models, integrity constraints and data manipulation operations, Relational query languages: Relational algebra, tuple and domain relational calculus.

Relational database design: Domain and data dependency, Armstrongs axioms, Normalization-1NF, 2NF, 3NF, BCNF, 4NF and 5NF, dependency preservation, lossless design.

Database Implementation: Introduction to SQL, DDL aspect of SQL, DML aspect of SQL update, insert, delete & various form of SELECT- simple, using special operators, aggregate functions, group by clause, sub query, joins, co-related sub query, union clause, exist operator. PL/SQL-cursor, stored function, stored procedure, triggers, error handling, and package.

Transaction Management and Concurrency control: Concept of Transaction, States of Transaction and its properties, Need of Concurrency control, concept of Lock, Two phase locking protocol.

Recovery Management: Need of Recovery Management, Concept of Stable Storage, Log Based Recovery Mechanism, Checkpoint.

Familiarization with standard: IEEE Standard Glossary of Software Engineering Terminology: IEEE 610.12-1990, POSIX® System Interfaces (relevant to database file handling and storage interaction): IEEE 1003.1,

Laboratory work: Database schema design, database creation, SQL programming: SQL commands to demonstrate the usage of DDL and DML, joining of tables, grouping of data and will implement PL/SQL constructs. They will also implement one project.

Project: It will contain database designing & implementation, should be given to group of 2-4 students. While doing projects emphasis should be more on back-end programming like use of SQL, concept of stored procedure, function, triggers, cursors, package etc. Project should have continuous evaluation and should be spread over different components

Course learning outcomes (CLOs):

On completion of this course, the students will be able to:

1. Analyze the Information Systems as socio-technical systems, its need and advantages as compared to traditional file-based systems.
2. Analyze and design database using E-R data model by identifying entities, attributes and relationships.
3. Apply and create Relational Database Design process with Normalization techniques.
4. Comprehend the concepts of transaction management, concurrence control and recovery management.

5. Demonstrate use of SQL and PL/SQL to implementation database applications.

Text Books:

1. Silverschatz A., Korth F. H. and Sudarshan S., Database System Concepts, Tata McGraw Hill (2010) 6th ed.
2. Elmasri R. and Navathe B. S., Fundamentals of Database Systems, Pearson (2016) 7th ed.

Reference Books and Supplementary Readings:

1. J. D. Ullman, Principles of Database Systems, Galgotia
2. https://cse.iitkgp.ac.in/~pabitra/course/dbms/dbms_new.html (Online Resources)
3. S. Feuerstein and B. Pribyl, Oracle PL/SQL Programming, 5th Edition, O'Reilly, 2009.